JUAN MAC DONAGH

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Profile

PhD candidate in structural bioinformatics (CONICET – National University of Quilmes), working on protein evolution, intrinsic disorder and conformational diversity. I combine evolutionary and structural analysis with programming and ML tools, while also being actively involved in teaching, open science and the scientific community.

Graduate and Postgraduate Education

National Scientific and Technical Research Council (CONICET) - National University of Quilmes, Argentina:
PhD fellow at the Structural Bioinformatics Group (SBG). Advisor: Prof. Dr. Gustavo Parisi **2022 – 2027**

Faculty of Exact Sciences - National University of La Plata:
Bachelor's Degree in Biotechnology and Molecular Biology. GPA: 7,68/10, Final thesis grade: 10/10 **2016 – 2021**

Publications

[∗] **Juan Mac Donagh**, Nicolas Vergesio, Andrea Aguilar, Rodrigo Nores, Antonio Lagares, Maria Silvina Fornasari, and Gustavo Parisi. *Transposable elements as evolutionary substrates of protein disorder in the human proteome*. Preprint. 2026. DOI: 10.64898/2026.06.12.731867. First author.

[∗] **Juan Mac Donagh**, Nahuel Escobedo, Tadeo Saldaño, Luciana Rodriguez Sawicki, Nicolas Palopoli, Sebastian Fernandez Alberti, Maria Silvina Fornasari, and Gustavo Parisi. "Revealing Missing Protein–Ligand Interactions Using AlphaFold Predictions". In: *Journal of Molecular Biology* (2024). DOI: 10.1016/j.jmb.2024.168852. First author.

[∗] **Juan Mac Donagh**, Abril Marchesini, Agostina Spiga, Maximiliano José Fallico, Paula Nazarena Arrías, Alexander Miguel Monzon, Aimilia-Christina Vagiona, Mariane Gonçalves-Kulik, Pablo Mier, and Miguel A. Andrade-Navarro. "Structured Tandem Repeats in Protein Interactions". In: *International Journal of Molecular Sciences* (2024). DOI: 10.3390/ijms25052994. First author.

[∗] Manjeet Kumar, Sushama Michael, Jesús Alvarado-Valverde, András Zeke, Tamas Lazar, Juliana Glavina, Eszter Nagy-Kanta, **Juan Mac Donagh**, Zsofia E Kalman, Stefano Pascarelli, Nicolas Palopoli, László Dobson, Carmen Florencia Suarez, Kim Van Roey, Izabella Krystkowiak, Juan Esteban Griffin, Anurag Nagpal, Rajesh Bhardwaj, Francesca Diella, Bálint Mészáros, Kellie Dean, Norman E Davey, Rita Pancsa, Lucía B Chemes, and Toby J Gibson. "ELM—the Eukaryotic Linear Motif resource—2024 update". In: *Nucleic Acids Research* (2024). DOI: 10.1093/nar/gkad1058.

[∗] **Juan Mac Donagh**, Diego Javier Zea, Guillermo Benitez, Cristian Guisande Donadio, Julia Marchetti, Nicolas Palopoli, María Silvina Fornasari, and Gustavo Parisi. *Evolutionary rates in human amyloid proteins reveal their intrinsic metastability*. Preprint. 2022. DOI: 10.1101/2022.09.07.506994. First author.

[∗] Tadeo Saldaño, Nahuel Escobedo, Julia Marchetti, Diego Javier Zea, **Juan Mac Donagh**, Ana Julia Velez Rueda, Eduardo Gonik, Agustina García Melani, Julieta Novomisky Nechcoff, Martín N. Salas, Tomás Peters, Nicolás Demitroff, Sebastian Fernandez Alberti, Nicolas Palopoli, Maria Silvina Fornasari, and Gustavo Parisi. "Impact of protein conformational diversity on AlphaFold predictions". In: *Bioinformatics* (2021). DOI: 10.1101/2021.10.27.466189.

Additional research and international experience

- **Johannes Gutenberg-Universität, Computational Biology and Data Mining Group**
  Marie Curie Fellow — Horizon 2020 "IDPFun2"
  5-month stay: March 2025 – August 2025
- **Johannes Gutenberg-Universität, Computational Biology and Data Mining Group**
  Marie Curie Fellow — Horizon 2020 "REFRACT"
  3-month stay: November 2022 – January 2023
  3-month stay: March 2024 – May 2024

- **European Molecular Biology Laboratory (EMBL), Heidelberg — Gibson Lab**
  Marie Curie Fellow — Horizon 2020 "IDPFun"
  3-month stay: February 2023 – May 2023

Conference Presentations

[∗] **Juan Mac Donagh** and Gustavo Parisi. "Transposable Element Insertions Contribute to the Emergence of Protein Disorder during Primate Evolution". In: *IDPfun2 Midterm Meeting / JIF UNQ*. Main talk. 2025.

[∗] **Juan Mac Donagh**, Miguel A. Andrade-Navarro, and Gustavo Parisi. "Identifying differential codon usage bias in protein domains with an automated pipeline." In: *XIV A2B2C 2024*. Poster. 2024.

[∗] **Juan Mac Donagh**, Diego Zea, and Gustavo Parisi. "Proteins that adopt condensate states occupy extreme and opposing positions within the spectrum of evolutionary rates in the human proteome". In: *XIV A2B2C 2024*. Poster. 2024.

[∗] **Juan Mac Donagh**, Diego Zea, Nicolas Palopoli, María Silvina Fornasari, and Gustavo Parisi. "Evolutionary analysis reveals emerging properties of the different forms that proteins can adopt". In: *RABE V*. Poster. 2023.

[∗] **Juan Mac Donagh**, Diego Zea, Gustavo Benítez, C.E. Guisande Donadio, J. Marchetti, Nicolas Palopoli, María Silvina Fornasari, and Gustavo Parisi. "Evolutionary rates in human amyloid proteins reveal their intrinsic metastability". In: *XIC A2B2C 2022*. Poster. 2022.

Awards and Grants

ORFG: Open Scholarship Seed Awards: "Open training material in Spanish for database curation". February 2024 (Member)

H2020-Marie Curie Rise. MSCA-Rise: "IDPFun2", 2025 – 2029 (Member)

PICT-2022: "Amyloidogenic proteins: an evolutionary and population-based approach to elucidate the sequence bases of their metastability". 2024 – present (Member)

PICT-2018. "Evolutionary trajectories, conformational diversity and structural divergence as foundations for the development of bioinformatics tools for protein study". 2018 – 2024 (Member)

H2020-Marie Curie Rise. MSCA-Rise: "IDPFun", 2017 – 2022 (Member)

Priority Research Program of the National University of Quilmes: Simulation of molecular processes of physicochemical and biological relevance. (Member)

Teaching experience

Teaching Assistant in Introduction to Chemistry and General Chemistry
National University of La Plata, Faculty of Exact Sciences. **03/2021 – present**

Conference attendance

1st AJI conference: Federal science at the end of the world (Organizer and vice president, Buenos Aires, Argentina 16–17/10/2025)

1st ML4NGP Training School: Protein aggregation, intrinsic disorder and phase separation in the era of machine learning. (Oporto, Portugal 19–21/4/2023)

EMBO Workshop: Visualizing biological data, VIZBI (EMBL Advanced Training Centre, Heidelberg, Germany. 28–31/3/2023)

Symposium of Spanish-speaking Bioinformatics and Computational Biology Students, SHE2Bioinfo. (virtual event, 5/6/2022)

Skills and Competencies

Programming languages: Extensive experience: R, Python, Julia, Bash. Intermediate experience: C, Unity.
Structural bioinformatics: AlphaFold2/3, ChimeraX, PyMOL, TM-score (structure prediction, comparison, analysis and visualization).

Sequence & evolution: multiple sequence alignment (MAFFT, MUSCLE); homology and profile search (BLAST suite, HMMER, HH-suite); evolutionary rates (Rate4Site); UCSC Genome Browser.
Intrinsic disorder: IUPred, MobiDB, PUNCH2-light.
Machine / deep learning: PyTorch, TensorFlow/Keras.
R ecosystem: tidyverse, Bioconductor (Biostrings, GenomicRanges), bio3d, ape, phangorn, RMarkdown/Quarto, Shiny.
Development & infrastructure: Git, Docker, LaTeX, HTML/CSS/JS, computing clusters and HPC, reproducible workflows (Snakemake).
Data curation & open science: ELM (Eukaryotic Linear Motif resource); DisProt.
Languages: English (near-native; TOEFL, 2022); Spanish (native); Portuguese (intermediate); German (basic).

Leadership experience

Co-founder and vice president of the Young Researchers Association (AJI).
Co-advisor and tutor for Facundo Sanchez Trapes undergraduate final thesis project (2023, National University of La Plata - Faculty of Exact Sciences)

Scientific Community Involvement

Volunteer in XIV Congress of the Argentine Association of Bioinformatics and Computational Biology (A2B2C, La Plata, Argentina, 07–08/11/2024).
Volunteer in the Argentine Regional Student Group in Bioinformatics (RSG, from 2022 to 2023).